Oil Related Stocks Analysis

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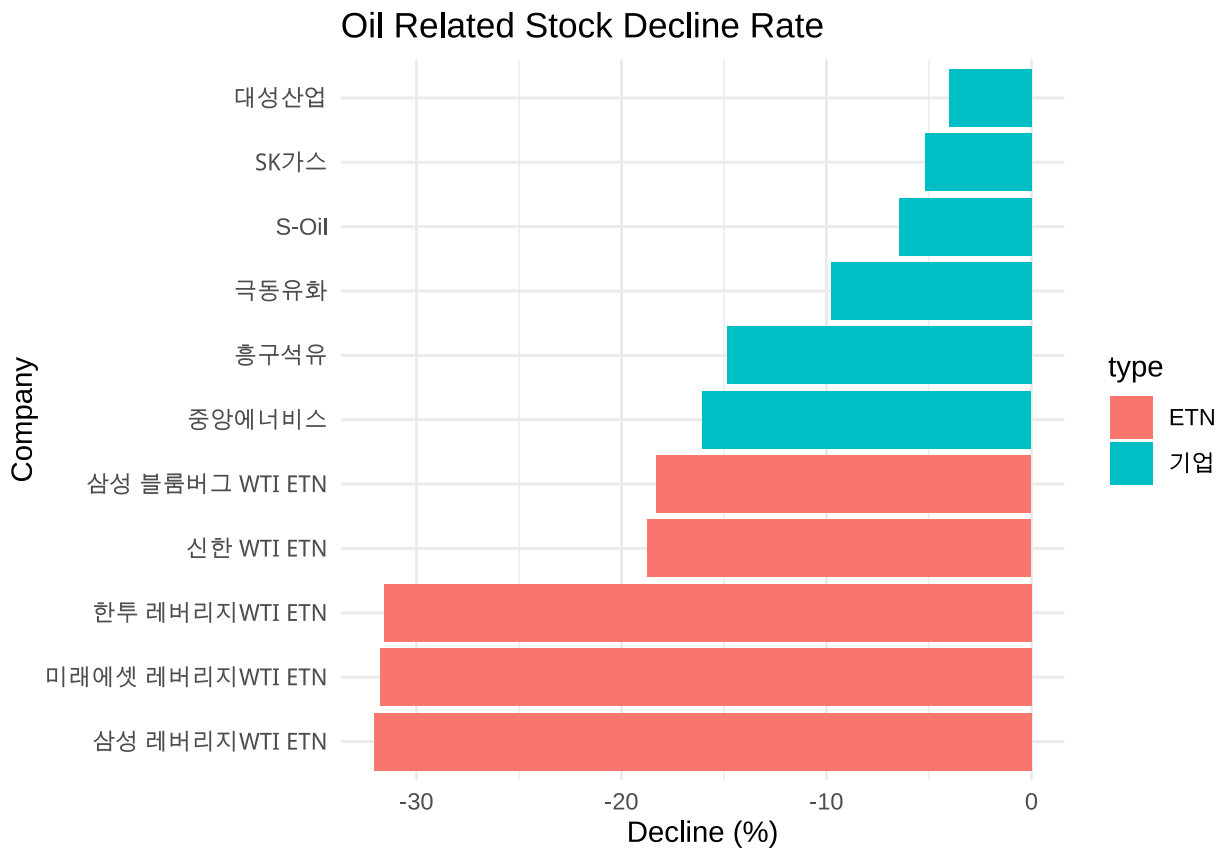
2026-03-13

```r
#1.
company <- c(
"S-Oil",
"    ",
"   ",
"   ",
"SK ",
"   ",
"     WTI ETN",
"     WTI ETN",
"      WTI ETN",
"   WTI ETN",
"      WTI ETN"
)

decline <- c(
-6.45,
-16.05,
-14.85,
-9.78,
-5.19,
-4.01,
-31.60,
-32.09,
-31.79,
-18.73,
-18.29
)

type <- c(
" "," "," "," "," "," ",
"ETN","ETN","ETN","ETN","ETN"
)

data <- data.frame(company, decline, type)

data
```

```
##                      company decline type
## 1                      S-Oil   -6.45
## 2                    -16.05
## 3                       -14.85
```

```
## 4                        -9.78
## 5                    SK    -5.19
## 6                        -4.01
## 7          WTI ETN  -31.60  ETN
## 8          WTI ETN  -32.09  ETN
## 9       WTI ETN  -31.79  ETN
## 10               WTI ETN  -18.73  ETN
## 11         WTI ETN  -18.29  ETN
```

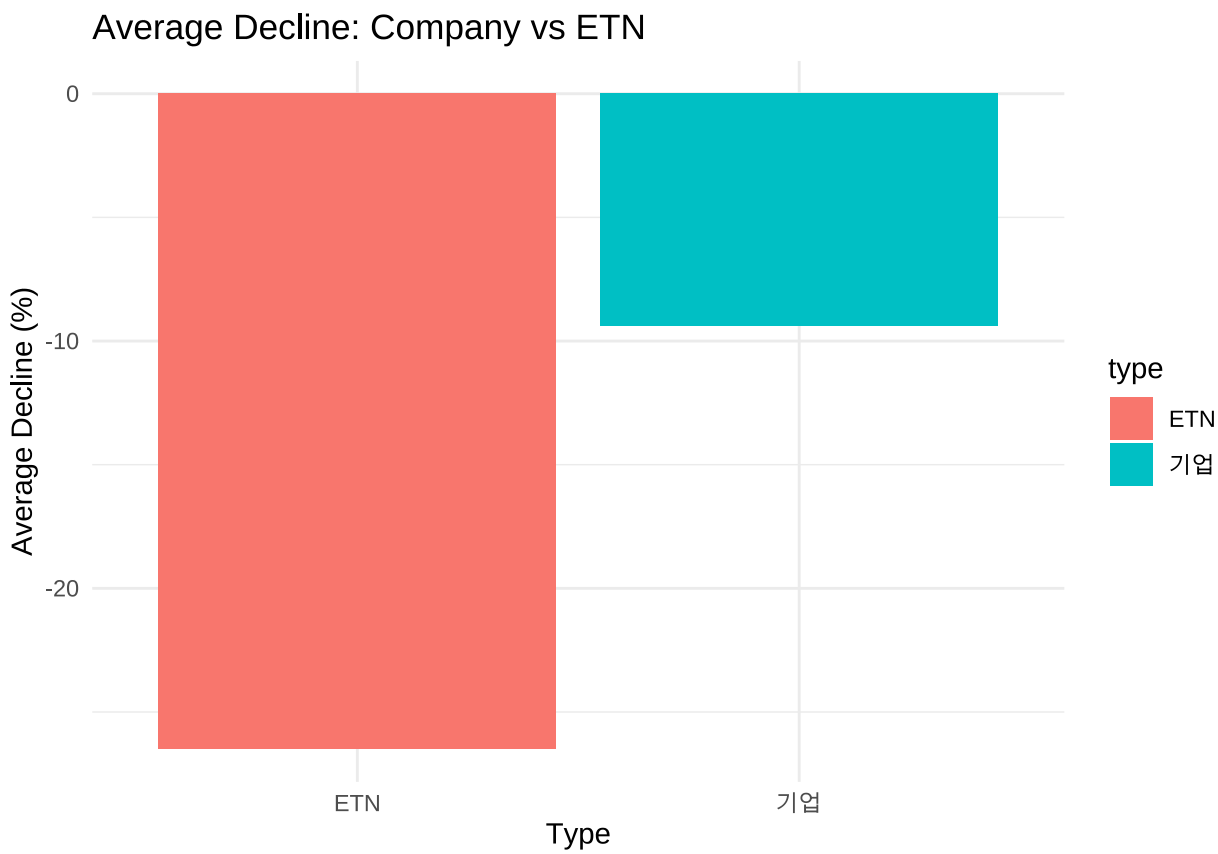
```
#2.
ggplot(data, aes(x=reorder(company, decline), y=decline, fill=type)) +
  geom_bar(stat="identity") +
  coord_flip() +
  labs(
    title="Oil Related Stock Decline Rate",
    x="Company",
    y="Decline (%)"
  ) +
  theme_minimal()
```




```
#3.   vs ETN
avg_data <- data %>%
  group_by(type) %>%
  summarise(avg_decline = mean(decline))

avg_data
```

```
## # A tibble: 2 x 2
##   type  avg_decline
##   <chr>       <dbl>
## 1 ETN        -26.5
## 2          -9.39
```

```r
ggplot(avg_data, aes(x=type, y=avg_decline, fill=type)) +
  geom_bar(stat="identity") +
  labs(
    title="Average Decline: Company vs ETN",
    x="Type",
    y="Average Decline (%)"
  ) +
  theme_minimal()
```




```r
#4. S-Oil
date <- c(
"2026-02-25",
"2026-02-26",
"2026-02-27",
"2026-03-03",
"2026-03-04",
"2026-03-05",
"2026-03-06",
"2026-03-09",
```

```r
"2026-03-10",
"2026-03-11"
)

price <- c(
112600,
113700,
110000,
141300,
126500,
134700,
129700,
128700,
117900,
114600
)

soil <- data.frame(date, price)
soil$date <- as.Date(soil$date)

soil
```

```
##          date  price
## 1  2026-02-25 112600
## 2  2026-02-26 113700
## 3  2026-02-27 110000
## 4  2026-03-03 141300
## 5  2026-03-04 126500
## 6  2026-03-05 134700
## 7  2026-03-06 129700
## 8  2026-03-09 128700
## 9  2026-03-10 117900
## 10 2026-03-11 114600
```

```r
#5. S-Oil
ggplot(soil, aes(x=date, y=price)) +
  geom_line(size=1) +
  geom_point() +
  labs(
    title="S-Oil Stock Price Trend",
    x="Date",
    y="Price (KRW)"
  ) +
  theme_minimal()
```

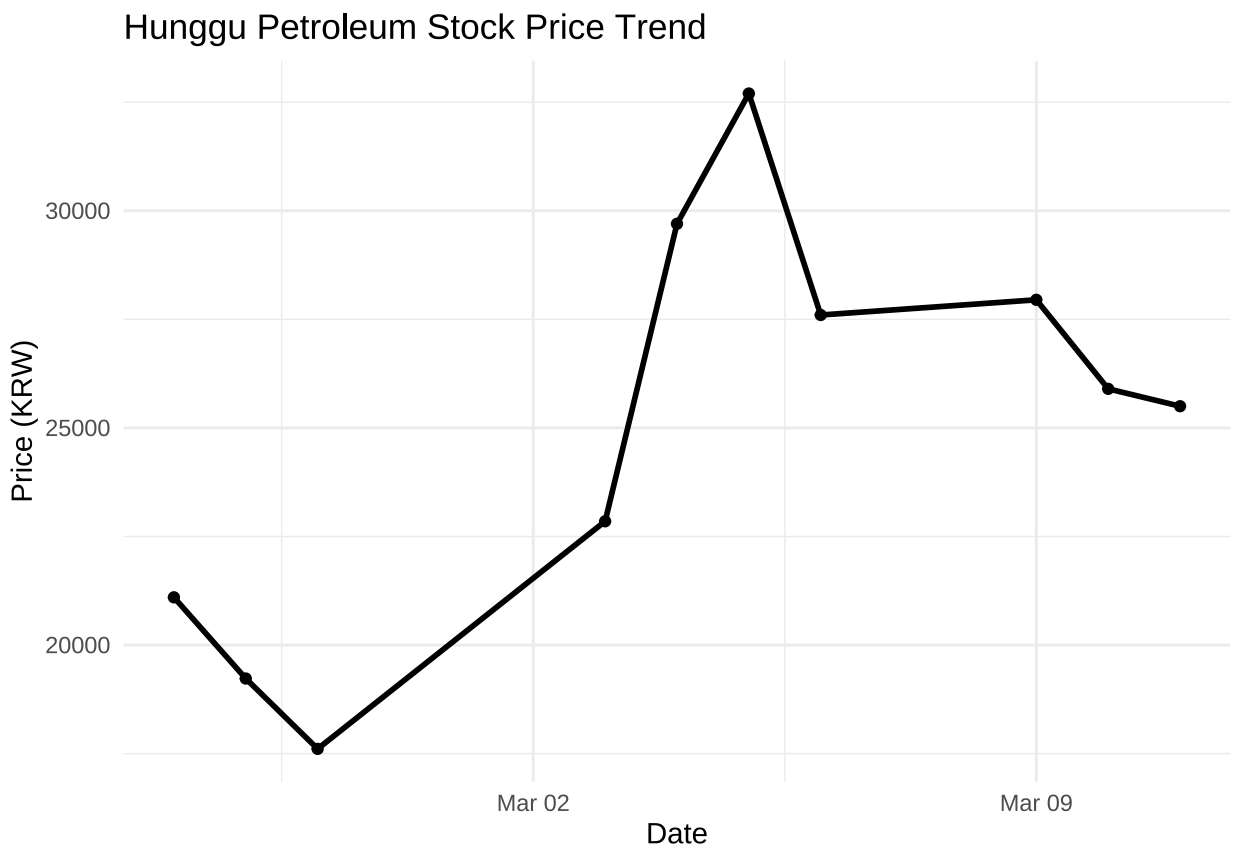
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
#6.
hung_date <- c(
"2026-02-25",
"2026-02-26",
"2026-02-27",
"2026-03-03",
"2026-03-04",
"2026-03-05",
"2026-03-06",
"2026-03-09",
"2026-03-10",
"2026-03-11"
)

hung_price <- c(
21100,
19230,
17610,
22850,
29700,
32700,
27600,
27950,
25900,
25500
)
```

```r
hung <- data.frame(hung_date, hung_price)
hung$hung_date <- as.Date(hung$hung_date)

hung
```

```
##     hung_date hung_price
## 1  2026-02-25      21100
## 2  2026-02-26      19230
## 3  2026-02-27      17610
## 4  2026-03-03      22850
## 5  2026-03-04      29700
## 6  2026-03-05      32700
## 7  2026-03-06      27600
## 8  2026-03-09      27950
## 9  2026-03-10      25900
## 10 2026-03-11      25500
```

```r
#7.
ggplot(hung, aes(x=hung_date, y=hung_price)) +
  geom_line(size=1) +
  geom_point() +
  labs(
    title="Hunggu Petroleum Stock Price Trend",
    x="Date",
    y="Price (KRW)"
  ) +
  theme_minimal()
```

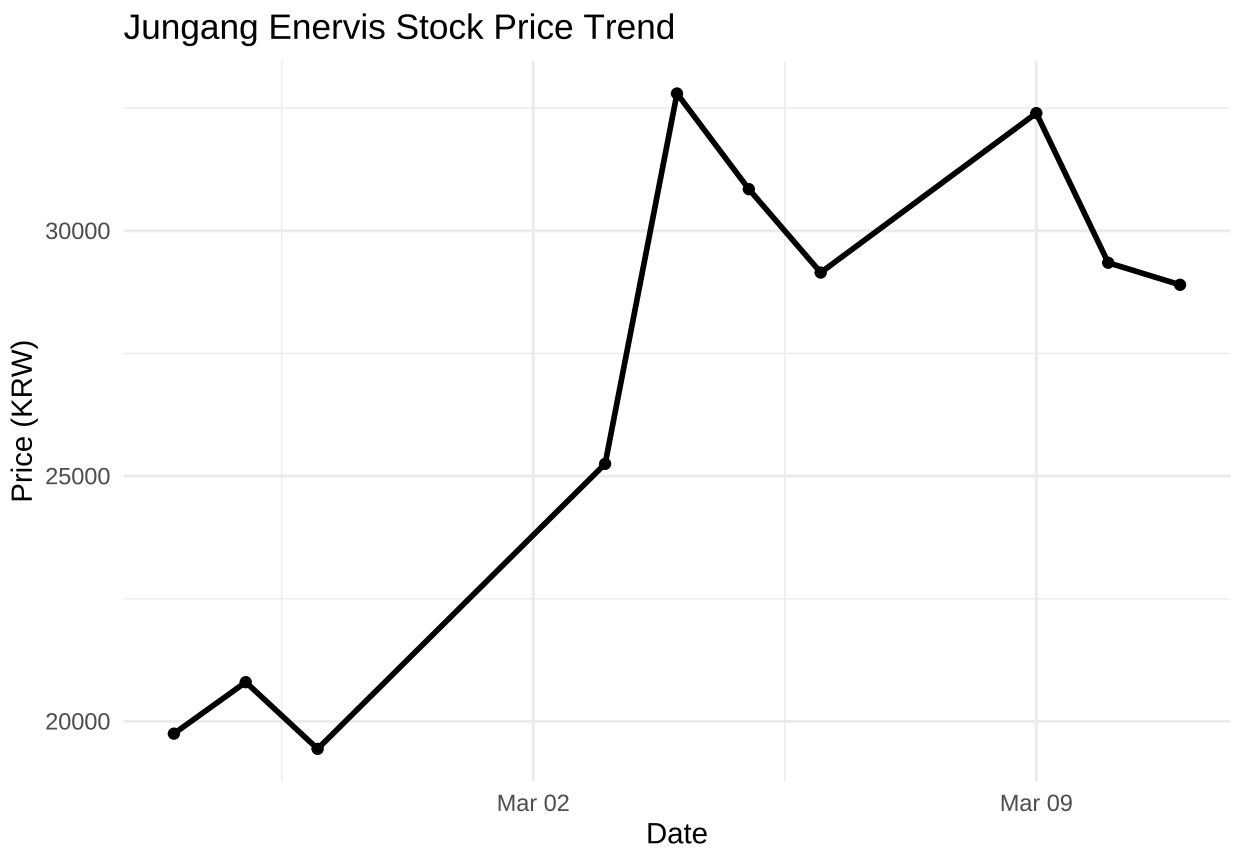



```
#8.
jung_date <- c(
"2026-02-25",
"2026-02-26",
"2026-02-27",
"2026-03-03",
"2026-03-04",
"2026-03-05",
"2026-03-06",
"2026-03-09",
"2026-03-10",
"2026-03-11"
)

jung_price <- c(
19750,
20800,
19440,
25250,
32800,
30850,
29150,
32400,
29350,
28900
)
```

```r
jung <- data.frame(jung_date, jung_price)

jung$jung_date <- as.Date(jung$jung_date)

jung
```

```
##     jung_date jung_price
## 1  2026-02-25      19750
## 2  2026-02-26      20800
## 3  2026-02-27      19440
## 4  2026-03-03      25250
## 5  2026-03-04      32800
## 6  2026-03-05      30850
## 7  2026-03-06      29150
## 8  2026-03-09      32400
## 9  2026-03-10      29350
## 10 2026-03-11      28900
```

```r
#9.
ggplot(jung, aes(x=jung_date, y=jung_price)) +
  geom_line(size=1) +
  geom_point() +
  labs(
    title="Jungang Enervis Stock Price Trend",
    x="Date",
    y="Price (KRW)"
  ) +
  theme_minimal()
```


Jungang Enervis Stock Price Trend
Price (KRW)
30000
25000
20000
Mar 02
Mar 09
Date